



A Perceptual Grouping Model for Auditory Kappa Effects

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Abstract

Current models suggest that perceptual time dilation between sequential auditory events with greater pitch separation, known as “auditory Kappa effects,” arise from auditory motion percepts. The prevailing explanation attributes this effect to violations of expected pitch trajectory, as the perceptual system imputes a constant pitch-velocity across events. We tested this *auditory motion hypothesis* against an alternative *auditory grouping hypothesis*, which proposes that pitch-timing interactions promote the grouping or segregation of auditory events, leading to subjectively shorter or longer perceived intervals between tones. We conducted two experiments to compare Kappa effects across sequences with varying pitch separations (velocities) and sequences with inconsistent pitch trajectories, which precluded stable pitch-velocity references. Experiment 1 results showed larger Kappa effects for sequences with slower pitch-velocity, contradicting the *motion hypothesis*, and Experiment 2 showed robust Kappa effects based on pitch proximity, regardless of unpredictable pitch motion. Phenomenological models associated with each hypothesis demonstrated that the *auditory grouping model* provided a better fit to our behavioral data in Experiment 2. These findings support perceptual grouping and segregation as fundamental processes underlying timing judgments and challenge our current understanding of feature interactions in auditory perception.

Keywords

Kappa effect; time perception; time dilation; auditory grouping; auditory motion

Introduction

Making sense of complex auditory environments requires the integration of various perceptual features such as pitch and onset timing into coherent objects and events (Bizley & Cohen, 2013). Theoretically dissociable acoustic cues produce reliable interactions in perception, as in the case of auditory Kappa and Tau effects, where greater pitch separation between successive tones leads to longer reported time intervals between them and longer time intervals between tones leads to greater reported pitch separation, respectively (Cohen et al., 1954; Henry & McAuley, 2009; Henry et al., 2009). In the aforementioned studies, two intervals were defined by three discrete events (termed ‘A-X-B’) and participants compared the duration of the first and second interval, reporting which is relatively longer

or shorter (between ‘A-X’ or between ‘X-B’). Kappa effects have been shown for visual, auditory, and tactile stimuli, and represent a general temporal bias where events separated by greater distance (e.g., in physical space or other feature space) are judged to be separated by longer time intervals (Cohen et al., 1953; Cohen et al., 1954; Suto, 1952).

Auditory Motion Hypothesis

The prevailing explanation for Kappa effects is an *auditory motion hypothesis*, which proposes that listeners attribute motion-like properties such as velocity and direction to these auditory patterns. This hypothesis is based on a broader theory of ‘phenomenological motion’ that links successive discrete events across space and time, leading to percepts of unified motion (Price-Williams, 1954). The foundational Kappa study in vision presented sequences of three flashes of light with varying distances between them, and participants reported longer durations between more spatially-distant flashes (Cohen et al., 1953; Huang & Jones, 1982). The distance between the first and third flash (‘A’ and ‘B’) was constant, but the location of the second (‘X’) flash was manipulated to be closer to either A or B. According to Jones and Huang (1982), participants impute uniform motion across sequences with an assumption of constant velocity and thus judge the farther X distance (from A or B) as taking longer to traverse than the shorter one, leading to longer duration judgments for the greater distance.

Based on other work relating properties of perceptual motion to pitch sequences (Eitan, 2013; Hubbard, 2017; Johnston & Jones, 2006), Jones and Huang’s imputed velocity model was adapted from the visuospatial to auditory pitch domain by computing a “pitch-velocity” (change in pitch over time) for tone sequences (Chen et al; 2016; Henry & McAuley, 2009). The auditory Kappa stimuli consisted of three sequential tones with fixed pitch distance (Δp_{AB}) between the first and last (‘A’ and ‘B’) tones (within a block of trials). The second (‘X’) tone varied from trial-to-trial to be closer to A or B in pitch. The *auditory motion hypothesis* put forth by Henry and McAuley (2009) proposes that the auditory Kappa effect arises due to imputing constant “pitch-velocity” across A-X-B tones, where all tones are expected to fall along a straight trajectory of pitch motion from A to B, leading the listener to judge a larger pitch distance between tones (i.e., Δp_{AX} or Δp_{XB}) as taking longer to traverse. Henry and McAuley further proposed that faster pitch-velocity should elicit stronger motion percepts, leading to larger Kappa effects (stronger pitch-based timing bias). Supporting the *auditory motion hypothesis*, they found that the imputed velocity model successfully predicted larger auditory Kappa effects for sequences with faster pitch-velocity when changing the denominator of $V = \Delta p_{AB} / IOI_T$, where Δp_{AB} is the total pitch distance between A and B tones and IOI_T is the total inter-onset time interval (IOI) between A and B.

A notable shortcoming of this hypothesis is the failure to adequately account for auditory Kappa effects in non-pitch features such as spectral centroid and speech vowel quality that may not induce auditory motion percepts (Marty et al., 2021; Shigeno, 1986). Marty et al. (2021) independently varied both pitch and brightness (spectral centroid) of tones in an A-X-B Kappa paradigm. They found Kappa effects for sequences of tones changing only in brightness with fixed pitch (fundamental frequency). Further, the magnitude of the brightness-only Kappa effect was not significantly different from sequences with both

features changing together (i.e., increasing in both pitch and brightness) and the pitch-only Kappa effect was significantly smaller than the other two conditions. This study showed that relative brightness can function as a stronger cue than pitch for Kappa effects, contrary to predictions of the *auditory motion hypothesis*.

In another study, Shigeno (1986) investigated Kappa effects for A-X-B sequences of vowel sounds, where A and B were brief synthesized /i/ and /e/ sounds, and X took on a range of intermediate sounds that did not belong to either phonetic category. Importantly, each possible target X sound was manipulated to be more similar to either the /i/ or /e/ sound. Results showed a Kappa effect where participants reported the presentation of X to be closer in time to the vowel sound that was more similar based on the first three formant frequencies. These experiments call into question the *auditory motion hypothesis* as an explanation for auditory Kappa effects, as spectral changes associated with timbral brightness or vowel quality are examples of cues for category boundaries rather than auditory motion.

Auditory Grouping Hypothesis

We propose an alternative *auditory grouping hypothesis*, which can account for Kappa effects based on pitch features and other categorical cues. This hypothesis proposes that two auditory stimuli more similar in a given feature dimension are more likely to be perceptually grouped as part of an auditory object and thus judged as occurring closer in time than a third, more distinct stimulus. This hypothesis is grounded in a robust literature on perceptual grouping, whereby discrete auditory events may be perceived as part of a single source or object (Bizley & Cohen, 2013). Previous studies have shown that participants overestimate durations containing a salient event boundary, such as a change in pitch or timbre in a sequence of tones, relative to sequences with no such change (Ongchoco et al., 2023). Pitch similarity and temporal proximity exemplify two Gestalt principles of organization in the auditory domain (Młynarski & McDermott, 2019). These studies propose that auditory grouping subserves a figure-ground organizational process, whereby events “belonging together” are perceptually grouped as part of an auditory object or figure, leading to a decrease in perceived temporal distance.

A closely related, but distinct, phenomenon is auditory streaming, in which sequential sounds may be perceptually organized into a single coherent stream or segregated into multiple streams depending on acoustic similarity and timing (Bregman & Campbell, 1971; van Noorden, 1975; Moore & Gockel, 2002). Streaming is influenced by the same acoustic dimensions that support perceptual grouping, such as pitch, spectral content, spatial location, and temporal proximity, whereby sounds that are more similar along a given dimension are more likely to be perceived as belonging to the same stream (for reviews, see Bizley & Cohen, 2013; Moore & Gockel, 2012; Snyder & Alain, 2007). However, grouping and streaming are not equivalent processes. Grouping reflects local, short-timescale organization of adjacent events, whereas streaming typically emerges from the accumulation of grouping cues over longer sequences. Accordingly, grouping cues need not result in full stream segregation, particularly for short sequences such as those used in the present study, but

may instead reflect early organizational processes that form a foundation for streaming (Młynarski & McDermott, 2019).

Consistent with these perceptual phenomena, grouping based on temporal patterns has been shown to affect the perceived duration of silent intervals between sounds (Sasaki et al., 2002), and gaps across perceptually distinct groups or streams are overestimated in duration relative to gaps within a group (Fitzgibbons et al., 1974; Thorpe & Trehub, 1989; Micheyl et al., 2010). When temporal grouping cues are ambiguous, such as in the case of nearly-equivalent inter-onset intervals between events, listeners may rely on more salient features (e.g., pitch) to group similar sounds and segregate dissimilar sounds. Beyond pitch and spectral content, Kappa effects have been demonstrated for variations in sound intensity (loudness) (Alards-Tomalain et al., 2013) and sounds presented sequentially to the same or different ear (Bausenhart & Quinn, 2018). Across these studies, more similar stimuli are consistently judged to be closer in time than more distinctive stimuli, suggesting that auditory Kappa effects arise from a general perceptual grouping mechanism rather than from a motion-prediction process per se.

The present study presents two experiments to test the *auditory motion hypothesis* and associated imputed velocity model against our *auditory grouping hypothesis* and model as a parsimonious explanation for auditory Kappa effects. Model fits were compared for both imputed velocity and auditory grouping models for both experiments. The first experiment revisits the pitch-velocity effects found previously (Henry & McAuley, 2009), testing whether Kappa effects are greater for sequences with faster pitch-velocity. According to the *auditory motion hypothesis*, Kappa effects are greater for faster pitch-velocity conditions which induce stronger motion percepts.

Experiment 1 tested predictions of these hypotheses by creating pitch-velocity conditions in the opposite manner. Across pitch-velocity conditions, sequence inter-onset intervals (IOIs) were constant and pitch distance was manipulated ($V = \Delta p_{AB}/IOI_T$). The *auditory motion hypothesis* predicts larger Kappa effects for sequences with larger overall pitch distance (faster pitch-velocity) than sequences with smaller pitch distance (Henry & McAuley, 2009). In contrast, the *auditory grouping hypothesis* predicts larger Kappa effects for sequences with smaller overall pitch distance due to stronger grouping cues among tones that are relatively closer in pitch (Bregman, 1990; Micheyl et al., 2010; Moore & Gockel, 2012). We fixed the magnitudes of target tone ('X') pitch shifts across velocity conditions with different A – B pitch distances (Δp_{AB}), making the relative movement of X toward A or B pitch greater for smaller Δp_{AB} (slower pitch-velocity) conditions, resulting in stronger pitch-grouping cues.

Experiment 2 modified the pitch-based Kappa stimuli from Experiment 1 to prevent formation of a stable velocity referent across tone sequences by introducing unpredictable changes in pitch motion direction. Each 3-tone ('A-X-B') sequence had a target 'X' pitch that was never between the pitch of 'A' and 'B'. Because 'X' pitch was always below that of 'A' or above 'B,' each sequence had either a downward-then-upward pitch motion trajectory, or an upward-then-downward trajectory. Sequences were presented in randomized order such that the pitch trajectory per-trial was unpredictable, and uniform motion would

not be imputed across the three tones. The *auditory motion hypothesis* predicts no Kappa effects for these inconsistent pitch trajectory sequences because these trajectories violate the required assumption of constant velocity. The *auditory grouping hypothesis* predicts Kappa effects based on pitch similarity because the target ‘X’ pitch was always closer to either ‘A’ or ‘B’ pitch.

Experiment 1: Kappa Effects with Pitch-Based Velocity Manipulation

Experiment 1 was carried out as a partial replication and extension of a previous study evaluating the imputed velocity model of pitch-based auditory Kappa effects (Henry & McAuley, 2009). This compared the magnitude of observed Kappa effects across tone sequences which varied in total pitch distance to create different pitch-velocities. In contrast to the Henry and McAuley study, the total duration (defined as total inter-onset time interval, IOI_T) between bounding tones (‘A and B’) was fixed across all conditions and pitch-velocity (V) was manipulated by varying the total pitch distance between tones A and B ($V = \Delta p_{AB}/IOI_T$). Henry and McAuley found larger auditory Kappa effects for sequences with faster pitch-velocity when changing the denominator of $V = \Delta p_{AB}/IOI_T$, however, the velocity effect cannot be disentangled from an overall temporal separation effect, as faster pitch-velocities were created by shortening gaps (IOIs) between tones to reduce IOI_T . Thus, pitch-velocity (V) and total duration between tones were perfectly correlated. We addressed this limitation by creating different pitch-velocity conditions by changing the numerator of $V = \Delta p_{AB}/IOI_T$, testing the prediction of the *auditory motion hypothesis* that the pitch-velocity of tone sequences drives the auditory Kappa effect. This hypothesis predicts larger Kappa effects for faster (i.e., larger pitch separation, Δp_{AB}) compared to slower velocity conditions. The *auditory grouping hypothesis* predicts the opposite pattern of results. If the Kappa effect is the result of grouping together tones that are relatively closer in pitch, Kappa effects will be larger for slower velocity (i.e., smaller pitch separation) conditions due to stronger grouping cues among tones that are relatively closer in pitch compared to faster velocity sequences with larger pitch leaps between tones (see: Figure 1C). We put forth an auditory grouping model as an alternative to the imputed velocity model from Henry and McAuley (2009), comparing model fits to our behavioral data.

Method

Participants. Twenty-four naïve participants were recruited from the University of Michigan (UM) and a community-wide health research recruitment database (<https://umhealthresearch.org>). To estimate sample sizes, we conducted an a-priori power analysis in G*Power to detect an effect size between $\eta^2 = .17$ and $.22$ with a repeated-measures ANOVA for one sample, with alpha level = $.05$, and power = $.8$ which showed a required sample of 16 – 24 participants. Seven participants were excluded from analysis for not completing the experiment task as instructed, reaching less than 80% correct on the easiest (suprathreshold) timing level (see details in Exclusion Criteria section below). The final sample included seventeen participants ($n = 14$ female, $n = 3$ male) between the ages of 19 and 37 years ($M = 24.1$, $SD = 5.1$ years). Thirteen participants reported having some formal music training, ranging from 2 to 16 years ($M = 8.1$, $SD = 4.2$ yrs). All participants underwent an audiogram and had pure-tone thresholds below 25 dB HL at

octave frequencies from 250 – 8000 Hz. All participants in our study were young healthy adults, thus results from both experiments may not be generalizable to a broader sample of different age groups or clinical populations. They were compensated at a rate of \$15 per hour for their participation. This study was approved by the Institutional Review Board - Michigan Medicine at the University of Michigan and all participants provided informed consent prior to participation.

Stimuli and Design. Stimuli were sequences of three 100 ms pure tones presented in increasing-pitch order (Figure 1A). Tones were presented in an ‘A-X-B’ paradigm, with a fixed temporal separation of 728 ms inter-onset interval (IOI_T) between bounding (A and B) tones. Bounding tones had a fixed pitch separation of 6, 8, 14, or 18 semitones (ST). This produced the four pitch-velocity conditions of 6 ST/728 ms, 8 ST/728 ms, 14 ST/728 ms, and 18 ST/728 ms (Figure 1C). Tone A pitch was always fixed at a C4 (261.63 Hz) and tone B pitches corresponding to each velocity condition were: F#4, (369.99 Hz), G#4 (415.30 Hz), D5 (587.33 Hz), and F#5 (739.99). Pitch-velocity conditions were chosen to match the exact condition showing the strongest Kappa effect in the previous study by Henry & McAuley (2009) (8 ST/728 ms) plus one slower (6 ST/728 ms) and two faster (14 ST/728 ms & 18 ST/728 ms) conditions. Trials were blocked within a pitch-velocity, and the order of pitch-velocity condition was counterbalanced across participants using a Latin-square design.

Experiment 1 implemented a 4 X 5 X 8 within-subjects design. Four pitch velocities (6 ST/728 ms, 8 ST/728 ms, 14 ST/728 ms, 18 ST/728 ms) were crossed with five target pitch levels (-2, -1, 0, +1, +2 ST) and eight target timing levels (+/- 4%, 8%, 12%, and 16%). Target pitch was set relative to the pitch midpoint (0 ST) between bounding tones and is represented as a signed ST deviation from this midpoint (- lower, + higher). Target tone timing was set relative to the sequence bisection point (364 ms), or half of the total inter-onset interval ($IOI_T/2$) between bounding tones and is represented as percentage deviation (- earlier, + later) from this bisection interval (Figure 1B).

In all velocity conditions, the target tone (‘X’) varied in pitch and timing from trial-to-trial in the same manner. Target tone pitch varied relative to the 0 ST (baseline) condition at the pitch midpoint between tones A and B (Figure 1A). This baseline condition corresponded to (a target pitch of) 3, 4, 7, or 9 ST between tones A and B for each of the 4 velocity conditions, respectively. All other target pitches were 1 or 2 semitones (ST) closer to the pitch of A or B, creating 5 target tone pitch conditions: -2, -1, 0, +1, +2 ST (from pitch midpoint). All pitch manipulations were selected to exclude same-note or octave-equivalent (i.e., 0 or 12 ST) pitch intervals.

Target tone onset timing was set relative to the sequence temporal bisection point ($IOI_T/2$ = 364 ms) and took on one of eight values (+/- 4, 8, 12, 16%), where negative values were early and positive values were late with respect to the bisection point (Figure 1B). For example, the onset of a target tone presented at -12% timing was 43.68 ms early with respect to the bisection point (i.e., 320.32 ms after A). Early target timings created a pattern of “short-long” intervals and late timings created a “long-short” pattern. Target timings were selected based on previous studies on auditory Kappa effects to include a range of IOIs that

span sub- to supra-threshold temporal discrimination for psychometric analyses (described below). All target timing and target pitch conditions were fully crossed and presented in randomized order once per experimental run, with 8 runs of 40 trials presented for each pitch-velocity condition. Thus, a total of 320 trials were presented for each pitch-velocity block.

Equipment and Procedure. Participants listened to ‘A-X-B’ tone sequences over Sennheiser HD 650 headphones while sitting at a computer terminal in a sound-attenuating booth. Stimuli were delivered and responses recorded via a MATLAB program written using the AFC package (Ewert, 2013). Their task was to listen to each three-tone sequence, then respond whether the pattern of time intervals between tones was “long-short” or “short-long” by clicking a labeled button on the computer monitor. Participants were explicitly told to ignore the pitch of the tones and only respond based on the timing. After passing a practice stage to ensure task comprehension (see Practice Phase section below), participants completed four blocks of 320 experimental trials, one for each pitch-velocity condition, for a total of 1280 trials. Each of the four blocks were separated by a mandatory break. After all experimental trials, participants completed a short survey about demographics, music training, and any strategies used to perform the experiment task. The entire experimental session lasted around 2 hours.

Exclusion Criteria. Our participant screening and exclusion criteria were based on performance at two stages:

Practice Phase: To ensure task comprehension, participants completed 20 practice trials in the two easiest conditions ($\pm 12\%$ and 16% timing offsets) requiring a minimum accuracy of 75% . If a participant failed the initial practice run, they were informed that they did not meet the cutoff and were allowed to repeat the practice. Failure to pass after three attempts resulted in exclusion from the study (with compensation for time spent).

Experimental Phase: Although all participants who completed experimental trials had passed the practice phase, we applied a second screening criterion: participants had to achieve 80% accuracy in the easiest ($\pm 16\%$) timing conditions – clearly perceptible levels of timing deviation where ceiling performance is expected – during the experiment. This served a function similar to ‘catch trials,’ ensuring that participants who demonstrated capability during practice maintained attention throughout the main task.

Data Analysis. Proportions of “long-short” responses were computed for each experimental condition for each participant, and psychometric functions (2-parameter sigmoid) were fit to individual data using a logit link function. Point of Subjective Equality (PSE) was calculated for each pitch level at each velocity by computing the value of x (time) when $y = .5$ (corresponding to 50% long-short responses), then PSEs were averaged across participants. The PSE represents the theoretical time interval where the target sound should be presented (relative to the temporal midpoint between bounding sounds) to produce two perceptually-equivalent inter-onset intervals (IOIs). Negative PSE values correspond to a leftward shift in the psychometric function and represent a bias to respond “long-short,” and positive PSE values correspond to a rightward shift and bias to

respond “short-long.” A PSE of 0 would indicate no systematic timing bias with equal 50% rate for “long-short” and “short-long” responses at the temporal bisection point. Following Henry and McAuley (2009), a relative Constant Error (CE) measure was calculated by subtracting PSE from the bisection IOI (half of the total sequence IOI), then converting to a percentage of the bisection IOI (Gescheider, 1985). Positive relative CE represents a bias to respond “long-short” more (corresponding to negative PSE) and vice-versa (see: Figure S1 in Supplemental Materials Appendix A). As a measure of discrimination threshold, we calculated just-noticeable difference (JND) as the semi-interquartile range. Effects of target pitch and pitch-velocity on CE and JND were analyzed using linear mixed effects models with the *lmer4* package (Bates et al., 2015) and *p*-values were obtained with the *lmerTest* package (Kuznetsova et al., 2017) in the *R* environment (R Core Team, 2024). As confirmatory analyses, Bayes factors were computed using *brms* (Bürkner, 2017) and *bridgesampling* (Gronau, et al., 2020) *R* packages, and are reported in Supplemental Materials Appendix B.

Transparency and Openness: This study was conducted in compliance with TOP guidelines. All data in Experiments 1 and 2 were collected in the years 2023 – 2024. We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. This study was not preregistered. All data are publicly available online at: https://osf.io/fcmv5/?view_only=e8c90fc1f0a04265a056373dc6defe5f.

Phenomenological Models

Imputed Velocity Model: The imputed velocity model was adapted by Henry and McAuley (2009) from the visual imputed velocity model (Jones & Huang, 1982) to the auditory pitch domain. This model assumes that observers impute uniform velocity (e.g., “pitch-velocity”) across sequential stimuli and actively generate expectations about when a sound event should occur. Expected time ($E(t)$) intervals are based on average pitch-velocity (V), $V = \Delta p_{AB}/IOI_T$, where Δp_{AB} is the total pitch separation (in semitones) and IOI_T is the total temporal separation (inter-onset interval) between ‘A’ and ‘B’ tones. Perceived onset timing of a target (‘X’) tone (t_p) is a weighted average of objective timing (t) and expected timing ($E(t)$). The relative contribution of t and $E(t)$ to the perceived timing (t_p) is set with a free weighting parameter, w_I .

$$t_p = w_I t + (1 - w_I) E(t) \quad (1)$$

If $w_I = 1$, perceived timing (t_p) is based solely on the objective timing t , and no Kappa effect is observed. If $w_I = 0$, t_p is based solely on the expected timing, where $E(t) = \Delta p_{AX}/V$. The value of w_I represents the extent of response bias in temporal judgments based on expected timing from imputed pitch-velocity, V , and negatively correlates with the magnitude of the Kappa effect. The imputed velocity model is used to predict the response probability, R , for a forced-choice task with binary response categories (i.e., “short-long” vs “long-short” timing patterns) by fitting a sigmoidal psychometric curve derived from a generalized logistic function. The equation includes the stimulus magnitude, in this case,

perceived timing (t_p) as well as a gain parameter, γ , and bias parameter, θ . The imputed velocity model equation is thus:

$$R = \frac{1}{1 + e^{-\gamma(t_p + \theta)}} \quad (2)$$

Where R is a response probability (i.e., probability of responding “long-short”) ranging from 0 to 1, which increases in a sigmoidal manner as perceived onset timing of the target tone (t_p) increases (i.e., the first time interval is perceived as longer than the second).

Auditory Grouping Model: To compare with the *imputed velocity model*, we developed a competing *auditory grouping model*, which differs in several important theoretical ways: first, this model does not assume that Kappa effects are the result of violated temporal expectations due to listeners imputing total sequence velocity over trials in an experiment. The *auditory grouping model* proposes that listeners perceptually group together auditory events based on feature similarity, leading to perceptual biases, rather than making predictions about when each target tone will occur on a trial-by-trial basis based on fixed pitch and temporal separation (pitch-velocity) across tone sequences. Thus, when temporal proximity among auditory events is ambiguous, participants are biased to report that two sounds more similar in another feature dimension, such as pitch, occurred closer together in time than a third, more distinct sound. This concept is similar to perceptual time dilation across auditory event boundaries, whereby durations containing an auditory event boundary are judged as longer than equivalent-duration sounds between boundaries (Micheyl et al., 2010; Ongchoco et al., 2023).

A second divergence from the *imputed velocity model* is that the *auditory grouping hypothesis* predicts larger Kappa effects for sequences with stronger grouping cues, regardless of total pitch separation over time (pitch-velocity). For successive sounds with closer *relative proximity* on a given feature dimension, there is an increased likelihood of being grouped as part of an auditory object. For tones varying in pitch, this predicts stronger biases of temporal proximity among sequential tones closer in pitch relative to other tones in a given sequence. Given identical temporal relations among sequential tones, the total pitch distance between the first and second tone when compared to the total pitch distance between the second and third tone will predict the grouping of the second tone with the first or third tone. If the second tone is relatively closer in pitch to the first than third tone, the *auditory grouping hypothesis* predicts that the second tone is more likely to be perceived as relatively closer in time to the first than third tone, regardless of pitch motion or velocity. Thus, larger magnitude Kappa effects are predicted for sequences with less relative pitch distance among tones, not for sequences with greater pitch-velocity.

Our implementation of the *auditory grouping model* has a few key changes to the formulation of the term t_p (perceived timing) from the *imputed velocity model* (Equation 2). Perceived onset timing of the target tone (t_p) is a function of the relative timing (t_r) and the relative pitch distance (S_{rel}) of the target tone to each neighboring tone. Relative timing (t_r) of the target tone (temporal proximity of the target to the first and last tone) is considered

based on the total inter-onset interval between the first and last tone (IOI_T) in addition to the absolute inter-onset time interval (t) between the target tone and first tone.

$$t_r = \frac{t - IOI_T/2}{IOI_T/2} \quad (3)$$

Relative feature distance (S_{rel}) (i.e., pitch distance) is a cue for grouping with the first or third tone, in addition to relative target timing (t_r). The relative pitch distance of the target tone is the difference in absolute pitch distance from the target ('X') to first (S_{AX}) and last (S_{XB}) tones over the total pitch distance ($S_T = \Delta p_{AB}$) across all tones in the sequence (e.g., the pitch distance between A and B).

$$S_{rel} = \frac{S_{XB} - S_{AX}}{S_T} \quad (4)$$

A free weighting parameter, w_2 , is included to represent the contribution of relative pitch distance (S_{rel}) to perceived relative target timing (t_p). The weighting parameter (w_2) represents the biasing effect of the to-be-ignored stimulus feature (i.e., pitch) on perceived timing.

$$t_p = t_r - (S_{rel}w_2) \quad (5)$$

The *auditory grouping model* importantly does not consider the expected timing of the target tone as a function of total pitch-velocity from previous trials to estimate perceived timing (Equation 3) in contrast to the *imputed velocity model* (Equation 1).

Fitting Models to the Data

To fit the *imputed velocity model* to their data, Henry and McAuley (2009) estimated parameter values for each participant and each velocity separately to map perceived target timing to response proportions according to their model function (Equation 2). They used a single value of γ , θ , and w_1 for each participant to generate predicted proportions of “long-short” responses, R , for each combination of target tone pitch and timing from their experiment. Allowing the three free parameters (γ , θ , and w_1) to vary, they iteratively selected values of w_1 to minimize the root mean squared error (RMSE) of approximation (average difference between model predicted and observed values) between the predicted and observed value of R . After fitting the imputed velocity model to their data, they reported the average best-fitting weights for each pitch-velocity condition.

To fit both models to our data, we closely followed the procedures described by Henry and McAuley (2009). First we fit the *imputed velocity model* to our data and compared our model parameters and RMSE to the previous study. Then, we compared estimated

parameters and fits between their imputed velocity and our auditory grouping model for our data in all three present experiments. Using a custom MATLAB script and the *fmincon* command to minimize RMSE between predicted and observed R values, we estimated participant-specific γ , θ , and w parameters per pitch-velocity condition. For both models, values of w were constrained between 0 and 1 but γ and θ were unconstrained. Initial parameter inputs were $\gamma = .01$, $\theta = 0$, and $w = .5$ for all model fitting. Best model parameter estimates and RMSEs are reported for each experiment below.

Results

Kappa Effects: To test for Kappa effects across pitch conditions in this experiment, a linear mixed-effects model was performed on the CE with target pitch (ST deviation from midpoint), pitch-velocity, and their interaction as fixed effects and participant as a random effect. The effect of target pitch on CE was significant, where CE increased by 2.33% on average for every semitone (ST) shift in target pitch (95% CI [1.80, 2.86], $p < .001$). Positive slopes indicate the presence of a Kappa effect, where participants showed greater bias to respond “long-short” when the target pitch was closer to the pitch of tone B than tone A (Figure 2). There was no effect of pitch-velocity on CE, however there was a significant interaction between target pitch and pitch-velocity ($\beta = -0.10$, 95% CI [-0.14, -0.05], $p < .001$), with the slope of target pitch and CE being largest for the slowest (6 ST/728 ms) pitch-velocity. Steeper slopes indicated a greater magnitude of Kappa effect for slower compared to faster pitch velocities. Average CE across all velocity and target pitch conditions was 2.41%, and a one-sample t-test against 0 was significant ($t(339) = 10.97$, $p < .001$, 95% CI [1.98, 2.84]). Thus, across all conditions, CE was significantly positive indicating a general “long-short” response bias.

Effects of Pitch-Velocity on Temporal Discrimination: A similar linear mixed-effects model on JND showed no effect of pitch but a significant effect of velocity on temporal discrimination threshold ($\beta = 0.47$, 95% CI [0.29, 0.66], $p < .001$), with the largest JNDs observed for the fastest velocity condition (18 ST / 728 ms), and no interaction between target pitch and velocity. Thus, discrimination thresholds (JNDs) were not affected by target pitch deviations but were affected by pitch-velocity condition. As pitch-velocity increased, a greater difference between the two durations is required to accurately report whether timing pattern was “long-short” or “short-long.” JNDs ranged from 18.82 to 27.97 ($M = 22.08$) across all conditions.

Model Fits: To test predictions of the *auditory motion hypothesis*, we first fit the *imputed velocity model* to our data and compared parameter estimates across pitch-velocity conditions. The imputed velocity model produced RMSE values ($M = .13$, $SD = .07$) in a range similar to the previous study (Henry & McAuley, 2009). To examine the effect of different pitch-velocities on model weights, we ran a repeated-measures ANOVA on w_I values for our best-fitting model, which showed no effect of pitch-velocity, ($F(3, 48) = .51$, $p = .68$, $\eta^2 = .02$). Mean w_I values ($\pm$ SD) for the slowest-to-fastest pitch velocities (6 ST/728 ms, 8 ST/728 ms, 14 ST/728ms, 18 ST/728 ms) were: .92 $\pm$.11, .94 $\pm$.04, .88 $\pm$.23, .88 $\pm$.23. Thus, w_I values, which represent the perceptual weighting of expected timing based on pitch-velocity against actual timing of tones, did not significantly differ

across pitch-velocity conditions in our experiment. Individual model fits are presented for both models in Figure S2 (panel A) of Supplemental Materials (Appendix C).

We then fit our *auditory grouping* model and compared RMSEs between the two competing models. For our grouping model, values of the weighting parameter w_2 represent the relative contribution of pitch proximity to judgments of temporal proximity. We conducted an ANOVA on best-fitting w_2 values across velocity conditions, which showed no effect of pitch-velocity ($F(3, 48) = .34, p = .80, \eta^2 = .01$). Mean w_2 values ($\pm$ SD) for slowest-to-fastest pitch-velocities (6 ST/728 ms, 8 ST/728 ms, 14 ST/728ms, 18 ST/728 ms) were: .06 $\pm$.03, .06 $\pm$.04, .07 $\pm$.06, .07 $\pm$.07. The overall fit for our auditory grouping model was comparable to the imputed velocity model, with a mean RMSE of .12 (SD = .03). To compare model fits, we then conducted a repeated-measures ANOVA on RMSE values, which showed that the effect of model bordered on statistical significance ($F(1, 16) = 4.37, p = .053, \eta^2 = .21$), no effect of velocity ($F(3, 48) = 1.31, p = .28, \eta^2 = .08$) and no interaction ($F(3, 48) = .69, p = .51, \eta^2 = .04$). Figure 3A – B shows observed and predicted CEs from each model for all conditions in Experiment 1. In addition, a Wilcoxon signed-rank test indicated that the RMSE for the *perceptual grouping model* (Median = 0.120) was not significantly different from that of the *imputed velocity model* (Median = 0.121), $z = 0.82, p = .413$. A paired t-test across all RMSE values similarly showed a marginal but non-significant difference ($p = .056$).

However, a clear advantage for the perceptual grouping model emerged in the k -fold cross-validation analysis. When predicting data from held-out test folds, the *perceptual grouping* model significantly outperformed the *imputed velocity* model. The mean RMSE for the grouping model was lower than the imputed velocity model by an average of 0.026. This difference in out-of-sample prediction error was highly significant (Paired t-test: $t_{(16)} = 4.70, p < .001$; Wilcoxon signed-rank test: $p < .001$). These results suggest that while both models can fit the training data reasonably well, the *perceptual grouping* model generalizes significantly better to unseen data, suggesting it might provide a more robust mechanistic account of the behavioral results (see Appendix D – I in Supplemental Materials for details).

Discussion

Experiment 1 results replicated the pitch-based auditory Kappa effect found in previous studies: participants showed greater bias to respond “long-short” when the target pitch level was higher compared to when it was 0 or lower. Thus, when the pitch of the target tone was closer to the pitch of tone B than tone A, participants judged the target to be closer in time to B than A. In contrast to previous findings, our results showing an interaction between target pitch and velocity demonstrated a larger Kappa effect for slower compared to faster velocities when pitch-velocity was manipulated by changing the magnitude of pitch separations while keeping the time interval between A and B (IOI_T) constant. This effect was in the opposite direction of previous results from the study by Henry and McAuley (2009), who found larger Kappa effects for faster pitch-velocities created by manipulating only IOI_T . Further, in this experiment average CE was positive overall, indicating a general bias toward responding that sequences had a “long-short” timing pattern. Additionally, temporal discrimination thresholds (JNDs) were generally

larger for faster velocities, indicating that participants were worse at discriminating whether the sequences were “long-short” or “short-long” for conditions with larger pitch jumps between tones.

Modeling results showed parameter estimates and fits to our data for the *imputed velocity model*/Ka comparable to previous studies. Fitting our auditory grouping model produced similar results, and statistical comparisons were inconclusive. The grouping model showed a slightly lower RMSE than the velocity model, but the difference did not rise to statistical significance when the model fitting methods of Henry and McAuley (2009) were applied. However, this comparison was significant when cross-validation methods were used (see: Supplemental Materials Appendix D – I). As an interim summary, Experiment 1 supported predictions of the *auditory grouping over auditory motion hypothesis* by showing that sequences with stronger pitch-based grouping cues (Figure 1C) led to stronger Kappa effects (Figure 2), despite having slower pitch-velocity.

Experiment 2: Kappa Effects with Inconsistent Pitch Trajectories

Experiment 2 tested competing *auditory motion* and *auditory grouping hypotheses* and associated models by examining whether the auditory Kappa effect is present for sequences with changing auditory motion trajectories. Similar to Experiment 1, stimuli were three-tone (‘A-X-B’) sequences with a fixed inter-onset time interval between tones A and B, and the same four “pitch-velocity” conditions created by varying the pitch distance between A and B. In contrast to Experiment 1, tone sequences did not have pitches presented in increasing-frequency order. Instead, all sequences in Experiment 2 included a change in pitch motion direction, from downward-to-upward or from upward-to-downward, by manipulating the pitch of target tone X from trial-to-trial such that it was never between the pitch of tones A and B (see schematic in Figure 4A). Within each pitch-velocity condition, these inconsistent pitch motion trajectories were presented in randomized order. Sequences with unpredictable changes in pitch motion direction should prevent listeners from forming expectations of constant pitch-velocity proposed to give rise to auditory Kappa effects. The imputed velocity model relies on an assumption of constant pitch-velocity across trials, where expected timing of sequential tones follows a linear pitch-time trajectory. Thus, the *auditory motion hypothesis* predicts no Kappa effect for these sequences with inconsistent pitch motion direction due to a lack of reliable velocity cues based on pitch/time trajectories across sequences. However, the *auditory grouping hypothesis* predicts a Kappa effect where participants perceptually group together two tones closer in pitch, and perceptually segregate the third, more pitch-distant tone, leading to biased timing judgments. As in Experiment 1, we report results of the imputed velocity model and our auditory grouping model and compare model fits to our data.

Method

Participants. Twenty-nine naïve participants recruited from the University of Michigan community took part in the experiment. Twelve participants were excluded from analysis due to accuracy below 80% in the easiest timing conditions, indicating a failure to complete the experiment task as instructed. The final sample included seventeen participants (n

= 9 female, $n = 8$ male) between the ages of 18 and 41 years ($M = 24.1$, $SD = 7.4$ years). Due to the similarity of design and analyses, sample sizes were estimated based on power analyses from Experiment 1. Fifteen participants reported having some formal music training, ranging from 2 to 11 years ($M = 7.7$, $SD = 3.0$ yrs).

Stimuli and Design. As in Experiment 1, stimuli were three-tone sequences in an ‘A-X-B’ paradigm where tone A and tone B were separated by 728 ms and B was always 6, 8, 14, or 18 ST higher than A, as in Experiment 1 (Figure 4A). In Experiment 2, however, the three tones were not in increasing-frequency order. Target (‘X’) tone pitches were always lower than tone A or higher than tone B, thus, introducing a change in the direction of pitch motion within each sequence to prevent the formation of expected constant pitch velocities. When the target pitch was lower than A (negative X pitch conditions), it was relatively closer to A than B pitch and when the target pitch was higher than B (positive X pitch), it was relatively closer to B than A. We have labeled conditions based on pitch-velocity for easy comparison with Experiment 1 stimuli (Figure 4A). Within each pitch-velocity condition, A and B had fixed pitch separation (same as Experiment 1). The sign of X indicates the direction of pitch change: a negative sign (–) signifies that X pitch moved lower than A, while a positive sign (+) signifies that X pitch moved higher than B.

Experiment 2 implemented an 8 X 8 within-subjects design. Eight relative target pitch conditions ($\pm 6, 8, 14, 18$ ST) were presented across eight target timing levels ($\pm 4\%$, 8% , 12% , and 16%). Relative target pitch conditions represent the number of ST between X and A minus the number of ST between X and B (i.e., how close X pitch is to A relative to B). Four conditions had X pitches below A (– ST) and four above B (+ ST). This created two types of pitch contours where 50% of target pitches introduced a change in pitch motion direction from decreasing-to-increasing frequency and the other half changed direction from increasing-to-decreasing (Figure 4A). Thus, when the pitch of X was below A (– ST), the pitch was relatively closer to A than B and vice-versa. Within each relative pitch condition, the pitch of X was roved by 1 ST. Timing conditions (Figure 1B) as well as overall pitch distances between tones A and B (pitch-velocities) were identical to those described in Experiment 1 (Figure 4C).

All absolute pitch values for each condition are shown in Figure 4B. Because tone A was fixed across all pitch conditions, we roved the pitch of X by 1 ST such that target pitches were not identical across all conditions. In the ± 6 ST conditions, X was 1 to 2 ST below A or above B, in the ± 8 ST condition, X was 2 to 3 ST below A or above B, in the ± 14 ST conditions, X was 3 to 4 ST below A or above B, and in the ± 18 ST conditions, X was 4 to 5 ST below A or above B (Figure 4C). As in Experiment 1, pitch manipulations excluded same-note or octave-equivalent relations among pitches presented within a sequence. There were 4 blocks of trials corresponding to ± 6 ST, ± 8 ST, ± 14 ST, and ± 18 ST relative pitch conditions with 10 runs of 32 trials in each block for a total of 320 trials per block. Each trial run had 1 presentation of each of the 4 possible target pitch and timing offset combinations presented in randomized order.

Equipment and Procedure. All equipment and procedures were identical to Experiment 1.

Data Analysis. All procedures for computing dependent variables were identical to Experiment 1. Effects of target pitch condition on CE and JND were analyzed using linear mixed effects models using the same *R* packages as in Experiment 1.

Fitting Models to the Data

To fit the *imputed velocity model* to our data in Experiment 2, we followed a similar procedure to Experiment 1, with one change to model constraints: possible values of expected timing of the target tone, $E(t)$, were limited between 0 and 728 ms ($0 < E(t) < IOI_T$). Without this constraint, the *imputed velocity model* produces expected target timing [$E(t)$] values that correspond to the target (second) tone occurring before the first or after the third tone. When target pitch values are extreme (i.e., $\Delta p_{AX} < 0$, $\Delta p_{AX} > V$), estimated target timing in the imputed velocity model is negative or greater than the total sequence IOI (i.e., $IOI_T < E(t) < 0$), where listeners expect the target tone before the first or after the third tone, respectively, despite the target always being the second tone. Without constraining values of $E(t)$ in this experiment, this model predicts the incorrect order of sequential tones (e.g., X-A-B) based on target pitch. This would require a perceptual sequencing error that should not occur for our range of IOIs between tones (minimum IOI = 305.76 ms) (Hirsh, 1959; Van Noorden, 1975). All procedures for fitting the *auditory grouping model* in Experiment 2 were identical to Experiment 1.

Results

Kappa Effects: To test for Kappa effects based on relative pitch condition among tones in sequences with inconsistent pitch trajectories, we first calculated the target relative pitch distance from X to A or B as the difference in absolute pitch distance (in ST) from A to X and from X to B ($|AX| - |XB|$), collapsing across pitch-velocity conditions. Values are more negative when X pitch is closer to A relative to B and more positive when X pitch is relatively closer to B. We then performed a linear mixed-effects analysis on constant error (CE) with target relative pitch condition (in ST) as a fixed effect and participant as a random effect. Target relative pitch strongly predicted CE, where CE increased by .22% for every 1 ST shift in target pitch closer to B than A (95% CI [0.18, 0.27], $p < .001$). Thus, there was a Kappa effect where participants showed greater bias to respond “long-short” (+ CE) when the X pitch was relatively closer to tone B pitch than tone A, and a “short-long” bias when X was closer to the pitch of A relative to B (Figure 5). Ignoring overall pitch-velocity, the magnitude of Kappa effects scaled with relative pitch, being larger when X was closer in pitch to one bounding tone relative to the other. Average CE across all conditions was .51%, and a one-sample t-test showed that this was not significantly different from 0 ($t(271) = 1.50$, $p = .14$, 95% CI [-0.16, 1.19]). For comparison with Experiment 1, we have reported additional linear-mixed effects analyses treating target pitch (ST deviation from midpoint) and pitch-velocity as separate factors in Supplemental Materials (Appendix E).

Effects of Pitch-Velocity on Temporal Discrimination: To test for effects of pitch-velocity conditions on temporal discrimination, we performed a linear mixed-effects model on JND with target relative pitch as a fixed effect and participant as a random effect. This showed a significant effect of relative pitch on JND ($\beta = 0.19$, 95% CI [0.11, 0.27], $p < .001$), where discrimination thresholds increased as relative pitch distance increased.

Average JNDs ranged from 17.21 to 25.98 ($M = 20.88$) for this experiment, in line with previous experiments.

Model Fits: Figure 6 (A & B) shows observed and predicted CEs from each model for all conditions in Experiment 2. We first estimated parameters for the *imputed velocity model* and calculated RMSE for the best-fitting model ($M = .2$, $SD = .14$). To examine the effect of different pitch-velocities on model weights, we ran a repeated-measures ANOVA on w_1 values for our best-fitting model, which showed no effect of pitch-velocity, ($F(3, 48) = .28$, $p = .84$, $\eta^2 < .001$), indicating that w_1 values, did not significantly differ across pitch-velocity conditions. Mean w_1 values ($\pm$ SD) for the slowest-to-fastest pitch velocities (6 ST/728 ms, 8 ST/728 ms, 14 ST/728ms, 18 ST/728 ms) were: .79 $\pm$.30, .76 $\pm$.3, .80 $\pm$.26, .72 $\pm$.31. Individual model fits are presented for both models in Figure S2 (panel B) of Supplemental Material (Appendix C).

We then fit our *auditory grouping model* and compared RMSEs between the two competing models. The overall fit for our *auditory grouping model* was better than the *imputed velocity model*, with a mean RMSE of .13 ($SD = .06$). For our grouping model, values of the weighting parameter w_2 represent the contribution of relative pitch distance to judgments of temporal proximity. We conducted an ANOVA on best-fitting w_2 values across velocity conditions, which showed no effect of pitch-velocity ($F(3, 48) = .08$, $p = .48$, $\eta^2 = .01$). Mean w_2 values ($\pm$ SD) for slowest-to-fastest pitch-velocities (6 ST/728 ms, 8 ST/728 ms, 14 ST/728ms, 18 ST/728 ms) were: .07 $\pm$.24, .09 $\pm$.24, .04 $\pm$.05, .11 $\pm$.24. To compare model fits, we then conducted an ANOVA on RMSE values for the two models, which showed a main effect of model type ($F(1, 16) = 16.4$, $p < .001$, $\eta^2 = .51$), no effect of velocity ($F(3, 48) = .38$, $p = .77$, $\eta^2 = .02$) and no interaction ($F(3, 48) = .20$, $p = .90$, $\eta^2 = .01$), indicating that the *auditory grouping model* was a significantly better fit to our data than the *imputed velocity model* in Experiment 2. This was confirmed with a Wilcoxon signed-rank test indicating that the RMSE for the *grouping model* (Median = 0.098) was significantly lower than that of the *imputed velocity model* (Median = 0.114), $Z = 3.40$, $p < .001$. Model fits with k-fold cross validation for Experiment 2 indicate the same results (see Appendix D – II in Supplemental Materials for details).

Discussion

The results of Experiment 2 showed a pitch-based auditory Kappa effect for sequences of tones with unpredictable (i.e., randomized) changes in pitch motion direction per trial. Across all conditions, the relative pitch distance between tones was a significant predictor of the observed timing bias. Participants responded that the timing pattern was “long-short” more when the target tone (X) was relatively closer in pitch to tone B than tone A, and responded “short-long” more when X was relatively closer in pitch to tone A, despite all sequences having inconsistent pitch trajectories. Modeling results showed that the imputed velocity model was a significantly worse fit to our behavioral data than the auditory grouping model. These results support the *auditory grouping hypothesis* over the *auditory motion hypothesis* as an underlying explanation of auditory Kappa effects.

General Discussion

The goal of the present study was to test the prevailing *auditory motion hypothesis* against an *auditory grouping hypothesis* as a more parsimonious explanation for auditory Kappa effects. Two experiments were conducted in which three sequential sounds (A-X-B) changed in pitch and participants reported the relative durations of the first (A-X) and second (X-B) inter-onset interval (IOI) by responding that the IOI pattern was “short-long” or “long-short.” Experiment 1 tested the prediction of the *auditory motion hypothesis* that tone sequences with faster pitch-velocity lead to greater Kappa effects than slower velocities, against the *auditory grouping hypothesis*, which predicts the opposite pitch-velocity effects. Crucially, different velocity conditions were created by manipulating overall pitch separation among tones, keeping the timing (temporal separation) identical across pitch-velocity conditions. This allowed us to isolate effects of pitch-velocity with the temporal structure held constant. Experiment 2 tested the *auditory grouping hypothesis* prediction of Kappa effects for tone sequences with inconsistent pitch trajectories with non-uniform pitch motion against predicted null results from the *auditory motion hypothesis*. Models associated with each hypothesis were fit to our data and compared.

The key finding from Experiment 1 was larger Kappa effects for sequences with slower velocities when pitch-velocity was manipulated by changing the pitch distance (rather than timing) between tones. Slopes relating temporal bias (CE) to target pitch were steeper for slower velocity compared to faster velocity conditions, contradicting the findings of Henry and McAuley (2009), which showed the opposite pattern of results when manipulating pitch-velocity by changing the temporal distance (rather than pitch) between tones. Increasing pitch-velocity by increasing pitch distance caused smaller kappa effects in our study, but increasing pitch-velocity by decreasing temporal distance caused larger kappa effects in the Henry and McAuley (2009) study. From the perspective of the *motion hypothesis*, these results appear incompatible as faster pitch-velocity is proposed to induce stronger motion percepts, however, these findings make sense from the perspective of the *grouping hypothesis*. In our Experiment 1, tones in slower velocity conditions (i.e., 6 ST/728 ms) had pitches that were closer together, providing stronger grouping cues than in faster velocity conditions with greater pitch separation (i.e., 18 ST/728 ms) which were less likely to be grouped together (Moore & Gockel, 2002; Vliegen et al., 1999). For example, in the slowest velocity condition (6 ST/728 ms) when target tone X was shifted 2 semitones (ST) closer to A, there was 1 ST between A-X and 5 ST between X-B, promoting grouping X with A instead of B. For the fastest velocity (18 ST/728 ms), however, the same 2 ST pitch shift toward A corresponded to 7 ST between A-X and 11 ST between X-B, providing weaker grouping cues among tones (van Noorden, 1975).

Sequential sounds belonging to the same group are perceived as occurring closer in time than sounds belonging to distinct groups (Thorpe & Trehub, 1989). Pitch similarity and temporal proximity may both serve as cues for grouping auditory events. Our experiment corroborates previous findings of larger Kappa effects for smaller temporal distances between tones by showing a similar pattern of larger Kappa effects for smaller pitch distances. Smaller pitch distances with equivalent time intervals between tones promotes more grouping than relatively larger pitch distances. Likewise, shorter time intervals with

equivalent pitch distance promotes more grouping than longer gaps between tones. In support of the *auditory grouping hypothesis*, these results indicate that Kappa effects arise from a balance of grouping and segregation processes that depend upon relations among sequence events, where the context of pitch or temporal spacing among auditory events may allow two tones to be heard as belonging to one group and a third, more distinct tone may be perceptually segregated.

Experiment 2 behavioral results provided further support for the *auditory grouping hypothesis* by showing Kappa effects based on relative pitch distance for tone sequences with inconsistent pitch trajectories. Similar to previous studies of pitch-based Kappa effects, participants were biased to report that the two A-X-B tones closer in pitch were closer in time. However, in Experiment 2, each sequence had non-uniform motion with a change in pitch-direction from downward-to-upward or from upward-to-downward that was randomized across trials to prevent stable velocity referents. This Kappa effect cannot be explained by violations of expected pitch from imputing constant velocity across sequences. Ignoring pitch-velocity, the relative distance between the pitch of the target tone and bounding tones significantly predicted the magnitude of Kappa effect. The degree to which the second (X) tone was relatively closer in pitch to the first (A) than the third (B) tone or closer to the third than the first tone strongly predicted the bias toward responding “short-long” or “long-short,” respectively.

Results of our model testing confirmed that the auditory grouping model was a better fit to our behavioral data than the imputed velocity model, especially for Experiment 2. Across both experiments, values of the weighting parameter, w_1 or w_2 , did not significantly increase for faster pitch-velocity conditions, in contrast to predictions of the *auditory motion hypothesis*. Compared to the Henry and McAuley (2009) study, our modeling results could not confirm that changing pitch-velocity by altering the total change in pitch over time (while keeping the total IOI constant) affected the perceptual weighting of the actual time interval relative to the expected time interval based on imputed pitch-velocity. Parameter estimates from our auditory grouping model produced similar null results for pitch-velocity conditions. For Experiment 1, the grouping model showed a slightly better fit than the velocity model but the difference did not rise to statistical significance following the model fitting methods of Henry and McAuley (2009). However, this comparison was significant when cross-validation methods were used (see: Supplemental Materials Appendix D – I). While both models can account for Experiment 1 results, the auditory grouping model can also account for Experiment 2 results, demonstrating more generalizability across experimental contexts.

One advantage of the grouping model is that it may be applied to other features with quantifiable similarity beyond pitch, opening avenues for future research on temporal biasing effects, regardless of auditory motion percepts. Comparing the two models’ generalizability to temporal effects of motion-irrelevant features could strengthen this dissociation. Current temporal coherence models suggest that cross-feature temporal alignment facilitates grouping (Krishnan et al., 2014; Lu et al., 2024; Shamma et al., 2011; Szabo et al., 2016; Teki et al., 2013). In contrast, our model considers the converse: how non-temporal grouping features (in this study, pitch) affect temporal judgments, highlighting

the importance of considering bi-directional interactions between dynamically-changing temporal and non-temporal features in auditory perception. One limitation of our model's predictive capability is that it may not generalize to event sequences with extremely small or large time or feature separations. Our study used a range of supra-threshold (Harris, 1952; König, 1957) pitch intervals between tones that spanned across segregation and integration windows, however, our inter-onset time intervals were longer than many studies examining pitch and timing interactions (van Noorden, 1975). Thus, our model should be tested for generalizability to sequences with very short IOIs or very large pitch changes that lead to weaker grouping or integration. Future versions of this model should additionally consider the role of cognitive effort and attention in perceptual grouping of auditory events (Boltz, 1998; Thompson et al., 2011; Sussman, 2017).

Taken together, our behavioral and modeling results challenge the existing *auditory motion hypothesis* in favor of the *auditory grouping hypothesis* as an explanation for auditory kappa effects. Our findings support a long-standing literature demonstrating perceptual grouping based on features such as pitch, temporal structure, spatial location, and spectral shape/content (Bizley & Cohen, 2013; Micheyl et al., 2013; Moore & Gockel, 2012; Viswanathan, et al., 2022; Vliegen et al., 1999). Auditory perceptual grouping is an inferential process that reflects internalized natural statistics of real-world sounds where correlations among acoustic features provide relevant information for identifying sounds. Gestalt principles of grouping based on similarity and proximity may underly humans' ability to group sequential sounds that belong together as part of a coherent stream and, conversely, segregate distinct auditory objects or streams (Chakrabarty & Elhilali, 2019; Mlynarski & McDermott, 2019). Sequential sounds belonging to the same group may be perceived as occurring closer in time than sounds belonging to distinct groups (Sasaki et al., 2002; Thorpe & Trehub, 1989). This phenomenon is couched in ecologically-relevant experience with natural sounds that have meaningful correlations, for example, sounds coming from a single source are more likely to have greater similarity or proximity than sounds from distinct sources. Established heuristic and Gestalt grouping principles may form a common neurological and cognitive basis for feature-based perceptual time illusions (i.e., dilation or contraction) in the presence of temporal ambiguities in event sequences of varying length or complexity.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

- Alards-Tomalin D, Leboe-McGowan LC & Mondor TA Examining auditory kappa effects through manipulating intensity differences between sequential tones. *Psychological Research* 77, 480–491 (2013). 10.1007/s00426-012-0438-8 [PubMed: 22538374]

- Barsz K Auditory pattern perception: The effect of tone location on the discrimination of tonal sequences. *Perception & Psychophysics* 50, 290–296 (1991). 10.3758/BF03206752 [PubMed: 1754370]
- Bates D, Maechler M, Bolker B, Walker S, Christensen RHB, Singmann H, ... & Bolker MB (2015). Package ‘lme4’. *Convergence*, 12(1), 2.
- Bausenhardt KM, Quinn KR (2018). On the interplay of visuospatial and audiotemporal dominance: Evidence from a multimodal kappa effect. *Attention, Perception, & Psychophysics* 80, 535–552. 10.3758/s13414-017-1437-z
- Bizley JK, & Cohen YE (2013). The what, where and how of auditory-object perception. *Nature Reviews Neuroscience*, 14(10), 693–707. 10.1038/nrn3565 [PubMed: 24052177]
- Boltz MG (1998). The processing of temporal and nontemporal information in the remembering of event durations and musical structure. *Journal of Experimental Psychology: Human Perception and Performance*, 24(4), 1087–1104. 10.1037/0096-1523.24.4.1087 [PubMed: 9706710]
- Bregman AS, Liao C, & Levitan R (1990). Auditory grouping based on fundamental frequency and formant peak frequency. *Canadian Journal of Psychology/Revue Canadienne de Psychologie*, 44(3), 400 – 413. 10.1037/h0084255
- Bregman AS, & Campbell J (1971). Primary auditory stream segregation and perception of order in rapid sequences of tones. *Journal of Experimental Psychology*, 89(2), 244–249. 10.1037/h0031163 [PubMed: 5567132]
- Bürkner P, (2017). “brms: An R Package for Bayesian Multilevel Models Using Stan.” *Journal of Statistical Software*, 80(1), 1–28. 10.18637/jss.v080.i01
- Cohen J, Hansel CEM, & Sylvester JD (1953). A new phenomenon in time judgment. *Nature*, 172(4385), 901–901. 10.1038/172901a0 [PubMed: 13111224]
- Cohen J, Hansel CEM, & Sylvester JD (1954). Interdependence of temporal and auditory judgments. *Nature*, 174(4431), 642–644. 10.1038/174642a0 [PubMed: 13203595]
- Chakrabarty D, & Elhilali M (2019). A Gestalt inference model for auditory scene segregation. *PLOS Computational Biology*, 15(1), e1006711. 10.1371/journal.pcbi.1006711 [PubMed: 30668568]
- Chen Y, Zhang B, & Kording KP (2016). Speed constancy or only slowness: What drives the kappa effect. *PLOS ONE*, 11(4), e0154013. 10.1371/journal.pone.0154013 [PubMed: 27100097]
- Eitan Z (2013). How pitch and loudness shape musical space and motion. In Tan SL, Cohen A, Cohen AJ, Lipscomb SD, & Kendall RA (Eds.), *The psychology of music in multimedia*, (pp. 165–191). Oxford University Press
- Ewert SD (2013). AFC – A modular framework for running psychoacoustic experiments and computational perception models. In: German Acoustical Society (DEGA) (Ed.), *Proceedings of the International Conference on Acoustics AIA-DAGA 2013* (pp. 1326–1329). Merano, Italy: German Acoustical Society (DEGA).
- Fitzgibbons PJ, Pollatsek A, & Thomas IB (1974). Detection of temporal gaps within and between perceptual tonal groups. *Perception & Psychophysics*, 16(3), 522–528. 10.3758/BF03198581
- Gescheider G, (1985). *Psychophysics: Method, theory, and application* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates.
- Gronau QF, Singmann H, Wagenmakers E, (2020). “bridgesampling: An R Package for Estimating Normalizing Constants.” *Journal of Statistical Software*, 92(10), 1–29. doi:10.18637/jss.v092.i10
- Harris JD (1952). Pitch Discrimination. *The Journal of the Acoustical Society of America*, 24(6), 750–755. 10.1121/1.1906970
- Hartmann WM (1983). Localization of sound in rooms. *The Journal of the Acoustical Society of America*, 74(5), 1380–1391. 10.1121/1.390163 [PubMed: 6643850]
- Henry MJ, & McAuley JD (2009). Evaluation of an imputed pitch velocity model of the auditory kappa effect. *Journal of Experimental Psychology: Human Perception and Performance*, 35(2), 551. 10.1037/0096-1523.35.2.551 [PubMed: 19331507]
- Henry MJ, McAuley JD, & Zaleha M (2009). Evaluation of an imputed pitch velocity model of the auditory tau effect. *Attention, Perception, & Psychophysics*, 71, 1399–1413. 10.3758/APP.71.6.1399
- Huang YL, Jones B (1982). On the interdependence of temporal and spatial judgments. *Perception & Psychophysics*, 32, 7–14. 10.3758/BF03204862 [PubMed: 7133950]

- Hubbard TL (2017). Momentum in music: Musical succession as physical motion. *Psychomusicology: Music, Mind, and Brain*, 27(1), 14–30. 10.1037/pmu0000171
- Hirsh IJ (1959). Auditory Perception of Temporal Order. *The Journal of the Acoustical Society of America*, 31(6), 759–767. 10.1121/1.1907782
- Johnston HM, & Jones MR (2006). Higher order pattern structure influences auditory representational momentum. *Journal of Experimental Psychology: Human Perception and Performance*, 32(1), 2–17. 10.1037/0096-1523.32.1.2 [PubMed: 16478322]
- Jones B, & Huang YL (1982). Space-time dependencies in psychophysical judgment of extent and duration: Algebraic models of the tau and kappa effects. *Psychological Bulletin*, 91(1), 128–142. 10.1037/0033-2909.91.1.128
- König E (1957). Pitch Discrimination and Age. *Acta Oto-Laryngologica*, 48(5–6), 475–489. 10.3109/00016485709126909 [PubMed: 13508204]
- Krishnan L, Elhilali M, & Shamma S (2014). Segregating complex sound sources through temporal coherence. *PLoS Computational Biology*, 10(12), e1003985. 10.1371/journal.pcbi.1003985 [PubMed: 25521593]
- Kuznetsova A, Brockhoff PB, & Christensen RHB (2017). lmerTest package: tests in linear mixed effects models. *Journal of Statistical Software*, 82(13).
- Marty N, Marty M, & Pfeuty M (2021). Relative contribution of pitch and brightness to the auditory kappa effect. *Psychological Research*, 85(1), 55–67. 10.1007/s00426-019-01233-y [PubMed: 31440814]
- Micheyl C, Hunter C, & Oxenham AJ (2010). Auditory stream segregation and the perception of across-frequency synchrony. *Journal of Experimental Psychology: Human Perception and Performance*, 36(4), 1029–1039. 10.1037/a0017601 [PubMed: 20695716]
- Micheyl C, Schrater PR, & Oxenham AJ (2013). Auditory frequency and intensity discrimination explained using a cortical population rate code. *PLoS Computational Biology*, 9(11), e1003336. 10.1371/journal.pcbi.1003336 [PubMed: 24244142]
- Młynarski W, & McDermott JH (2019). Ecological origins of perceptual grouping principles in the auditory system. *Proceedings of the National Academy of Sciences*, 116(50), 25355–25364. 10.1073/pnas.1903887116
- Moore BC, & Gockel H (2002). Factors influencing sequential stream segregation. *Acta Acustica United with Acustica*, 88(3), 320–333.
- Moore BC, & Gockel HE (2012). Properties of auditory stream formation. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367(1591), 919–931. 10.1098/rstb.2011.0355
- Lu K, Dutta K, Mohammed A, Elhilali M, & Shamma S (2024). Temporal-coherence induces binding of responses to sound sequences in ferret auditory cortex. *bioRxiv*, 2024–05. 10.1101/2024.05.21.595170
- Ongchoco JDK, Yates TS, & Scholl BJ (2023). Event segmentation structures temporal experience: Simultaneous dilation and contraction in rhythmic reproductions. *Journal of Experimental Psychology: General*, 152(11), 3266. <https://psycnet.apa.org/doi/10.1037/xge0001447> [PubMed: 37768575]
- Price-Williams DR (1954). The kappa effect. *Nature*, 173(4399), 363–364. [PubMed: 13144767]
- R Core Team (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <<https://www.R-project.org/>>
- Sasaki T, Suetomi D, Nakajima Y, & Ten Hoopen G (2002). Time-shrinking, its propagation, and Gestalt principles. *Perception & psychophysics*, 64(6), 919–931. 10.3758/BF03196796 [PubMed: 12269299]
- Shamma SA, Elhilali M, & Micheyl C (2011). Temporal coherence and attention in auditory scene analysis. *Trends in Neurosciences*, 34(3), 114–123. 10.1016/j.tins.2010.11.002 [PubMed: 21196054]
- Shigeno S (1986). The auditory tau and kappa effects for speech and nonspeech stimuli. *Perception & Psychophysics*, 40, 9–19. 10.3758/BF03207588 [PubMed: 3748768]
- Snyder JS, & Alain C (2007). Toward a neurophysiological theory of auditory stream segregation. *Psychological Bulletin*, 133(5), 780. 10.1037/0033-2909.133.5.780 [PubMed: 17723030]

- Sussman ES (2017). Auditory Scene Analysis: An Attention Perspective. *Journal of Speech, Language, and Hearing Research*, 60(10), 2989–3000. 10.1044/2017_JSLHR-H-17-0041
- Suto Y (1952). The effect of space on time estimation (S-effect) in tactual space. *Japanese Journal of Psychology*, 22(3), 189–200.
- Szabó BT, Denham SL, & Winkler I (2016). Computational models of auditory scene analysis: A review. *Frontiers in Neuroscience*, 10, 524. 10.3389/fnins.2016.00524 [PubMed: 27895552]
- Teki S, Chait M, Kumar S, Shamma S, & Griffiths TD (2013). Segregation of complex acoustic scenes based on temporal coherence. *Elife*, 2, e00699. 10.7554/eLife.00699 [PubMed: 23898398]
- Thompson S, Carlyon R, & Cusack R (2011). An Objective Measurement of the Build-Up of Auditory Streaming and of Its Modulation by Attention. *Journal of Experimental Psychology. Human Perception and Performance*, 37, 1253–1262. 10.1037/a0021925 [PubMed: 21480747]
- Thorpe LA, & Trehub SE (1989). Duration illusion and auditory grouping in infancy. *Developmental Psychology*, 25(1), 122. 10.1037/0012-1649.25.1.122
- van Noorden LPAS (1975) Temporal coherence in the perception of tone sequences. Institute for Perception Research, Technical Univ. Eindhoven. 10.6100/IR152538
- Viswanathan V, Shinn-Cunningham BG, & Heinz MG (2022). Speech Categorization Reveals the Role of Early-Stage Temporal-Coherence Processing in Auditory Scene Analysis. *Journal of Neuroscience*, 42(2), 240–254. 10.1523/JNEUROSCI.1610-21.2021 [PubMed: 34764159]
- Vliegen J, Moore BC, & Oxenham AJ (1999). The role of spectral and periodicity cues in auditory stream segregation, measured using a temporal discrimination task. *The Journal of the Acoustical Society of America*, 106(2), 938–945. 10.1121/1.427140 [PubMed: 10462799]

Public Significance Statement

Accurately perceiving the timing of sequential sounds is essential for speech and music perception. Past research has suggested that non-temporal features like pitch can influence how we perceive the timing of sequential sounds by creating a perception of auditory motion. This study supports an alternative explanation for this perceptual bias based on auditory grouping, which explains a broader range of timing biases, including those not accounted for by auditory motion.

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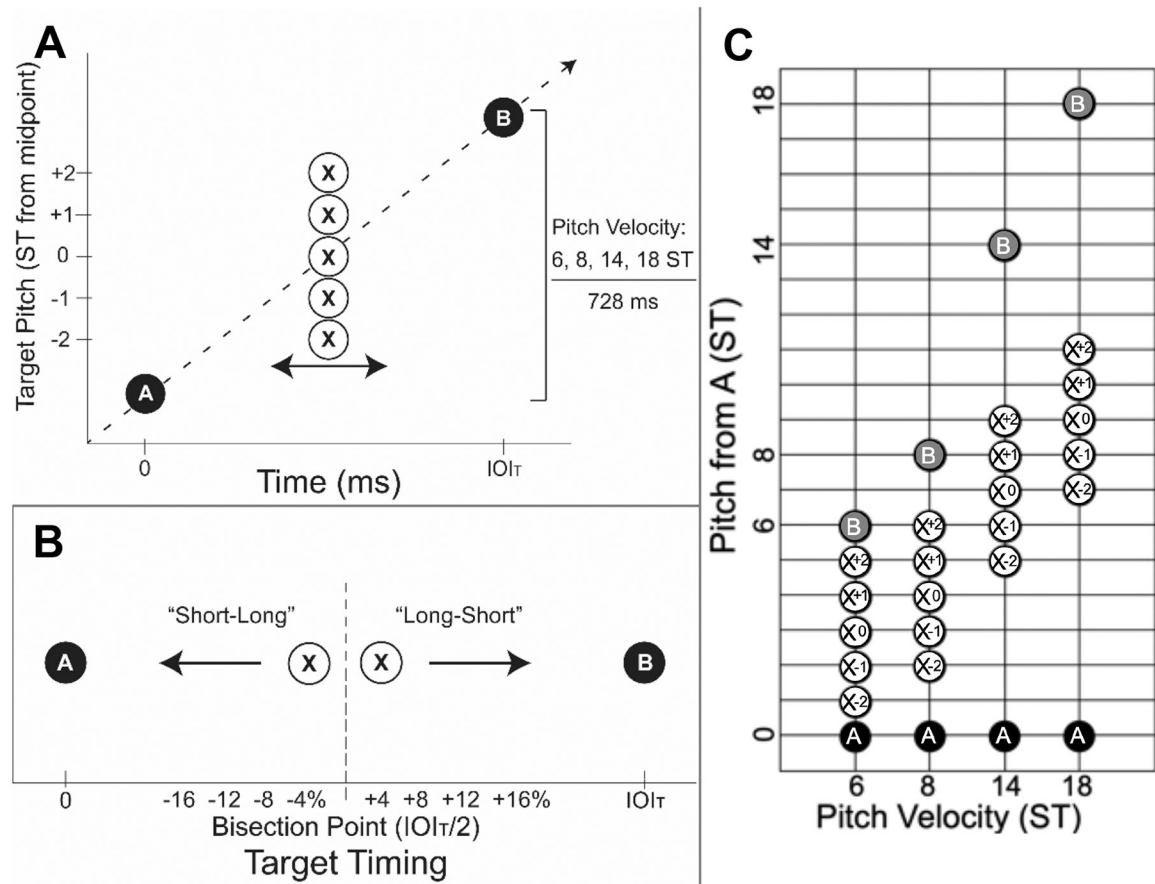


Figure 1.

Experiment 1 stimulus schematics. **A**) Five possible target ('X') pitch conditions. 'A' and 'B' had fixed pitch and temporal separation per block. Target pitch 'X' (open circles) varied between 0 (pitch midpoint) and 2 semitones (ST) higher and lower from trial-to-trial. Four pitch-velocity conditions (calculated as total pitch distance in semitones over total time) were created by changing the pitch separation between A and B, keeping the time interval between them (IOI_T) constant. Dashed line represents the expected pitch trajectory according to the imputed pitch-velocity model (Henry & McAuley, 2009). **B**) *Right*: Eight target timing conditions for all experiments. The target tone was presented early (-) or late (+) with respect to the temporal bisection point (IOI_T/2) by 4, 8, 12 or 16%. Early targets created "short-long" IOI patterns, and late targets created "long-short" patterns. **C**) Illustration of all possible target pitches in the context of each pitch-velocity condition. According to the grouping hypothesis, a 1 or 2 ST shift in X toward A or B is predicted to be a stronger grouping cue with the closer pitch for the 6 ST compared to 18 ST condition.

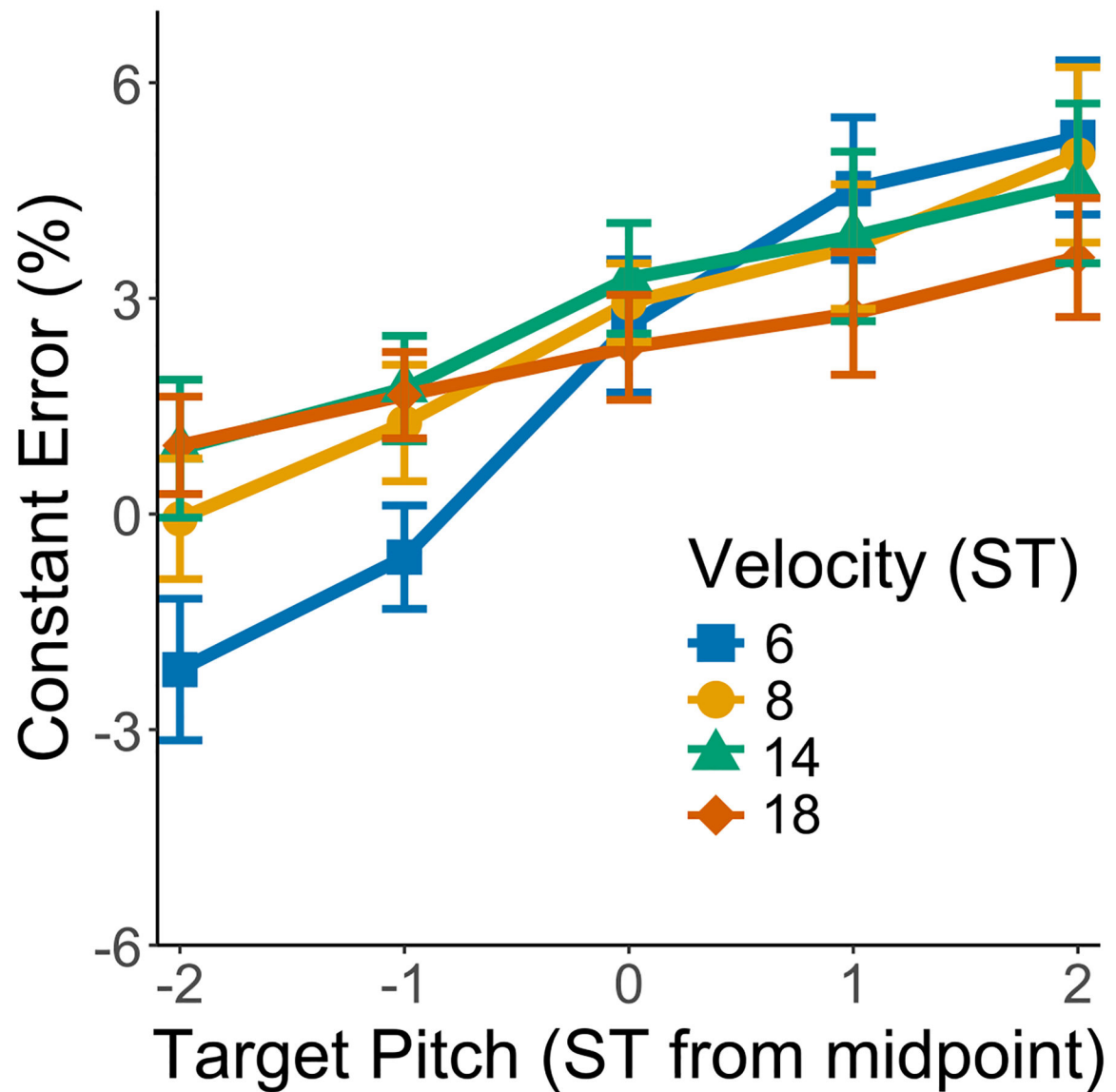
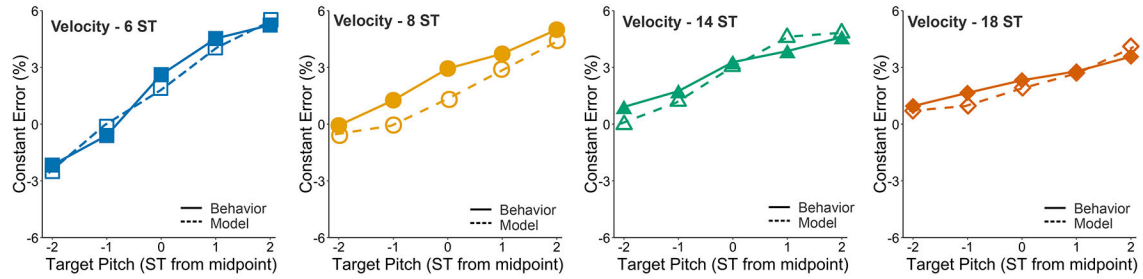
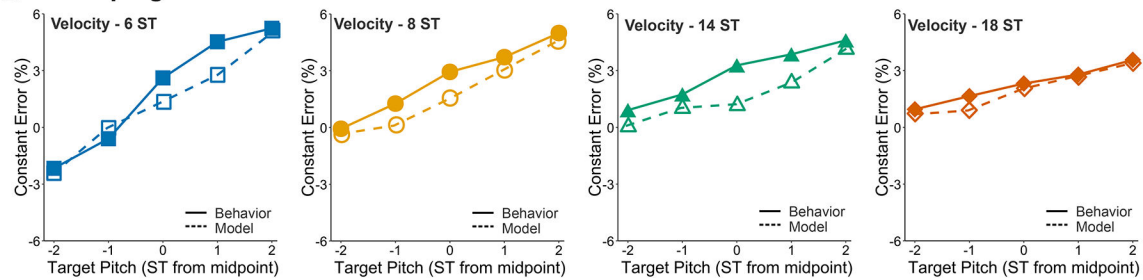


Figure 2.

Experiment 1 results showing mean CE with standard error bars as a function of target pitch in the four pitch-velocity conditions. Kappa effects were larger (steeper slopes) for slower (e.g., 6 ST) compared to faster (e.g., 18 ST) velocity conditions.

A Imputed velocity model**B Grouping model****Figure 3.**

Average CE from Experiment 1 showing observed data in solid lines with filled markers and model-predicted data in dashed lines with open markers. **A)** Imputed velocity model. **B)** Perceptual grouping model. After calculating best-fitting model parameters for each subject and pitch-velocity, respective models were used to calculate predicted proportion of “long-short” responses (Equation 2) and psychometric measures were estimated following the same procedures for observed and model-predicted data (as detailed in Method section).

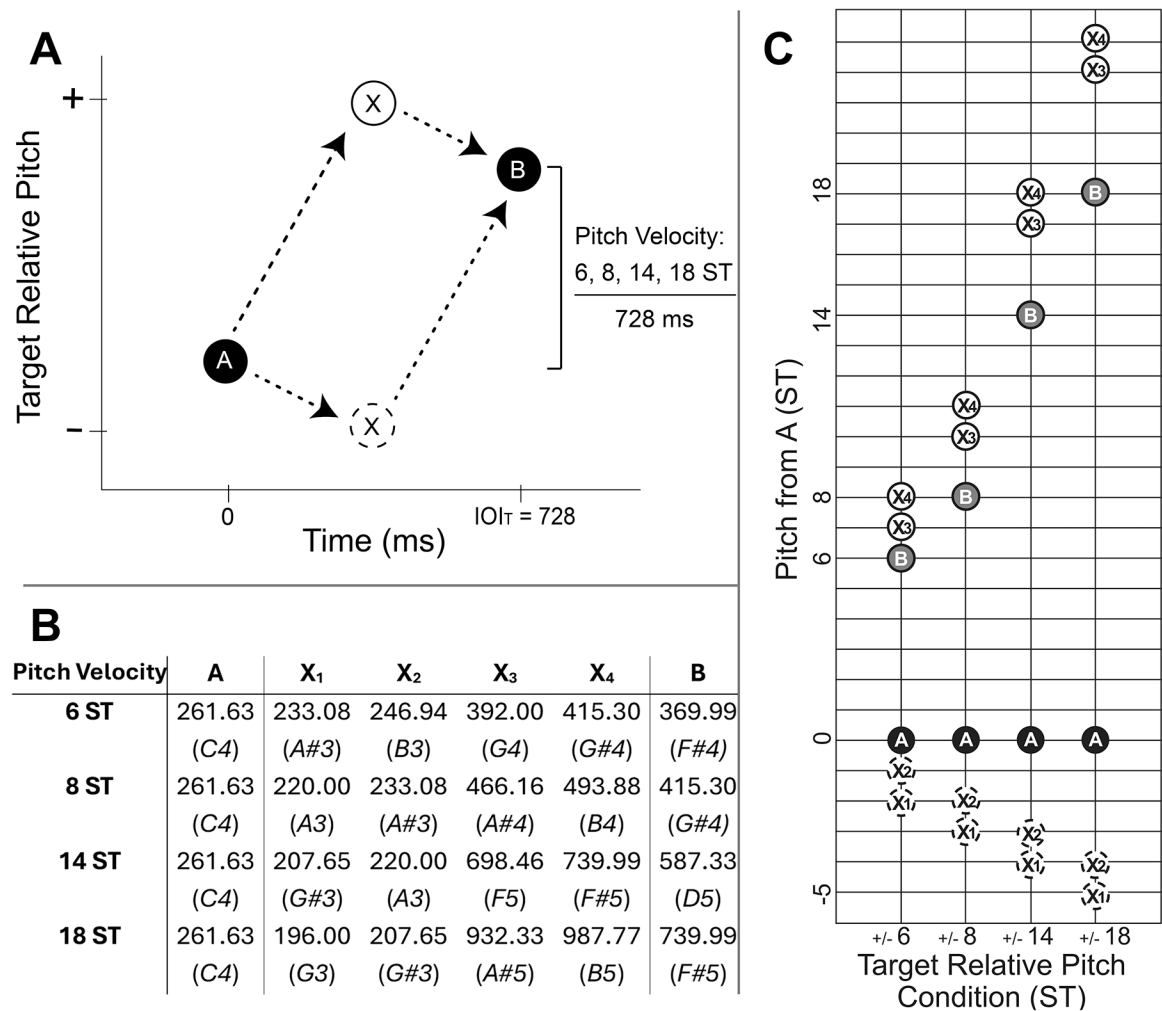


Figure 4. Exp 2 stimulus schematics. **A)** Inconsistent pitch trajectories formed by target ‘X’ pitches that were never between A and B. Pitch-velocities were the same as Exp 1, but all sequences included a change in motion direction. **B)** Table of all tone frequencies in Hz (musical note names in parentheses). Rows show all possible pitch values with roving. Tone A was fixed at 261.63 Hz and tone B was 6, 8, 14, or 18 semitones (ST) higher than A. Target tone X pitch varied and was either below A or above B. X₁ and X₂ are roved values of X pitches closer to A (dashed circles in panels A & C), X₃ and X₄ are pitches closer to B. **C)** Illustration of all possible target pitches roved in each relative pitch condition. Negative pitches (dashed circles: X₁, X₂) represent X pitch closer to (lower than) A and positive (solid circles: X₃, X₄) represent X pitch closer to (higher than) B.

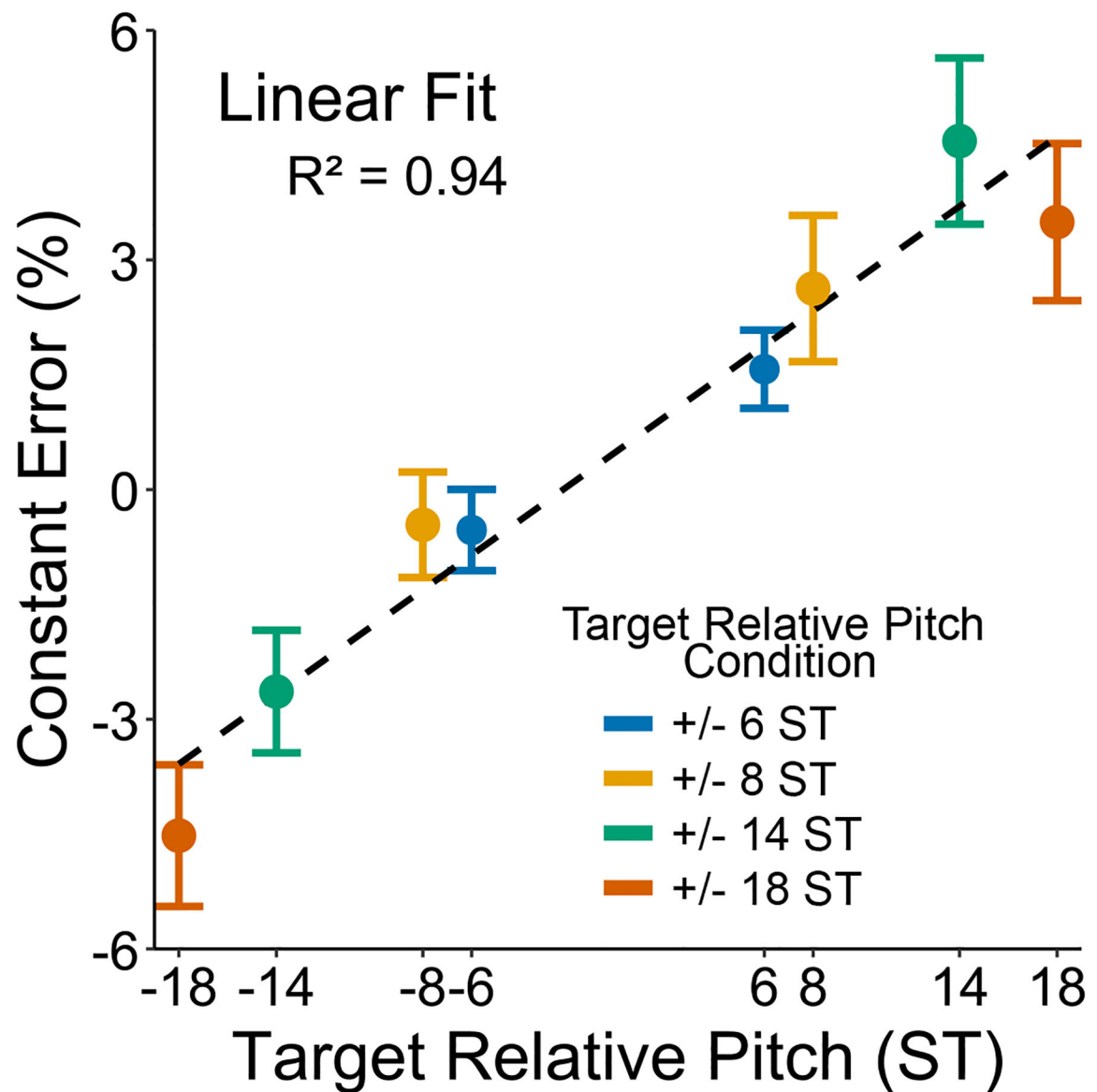
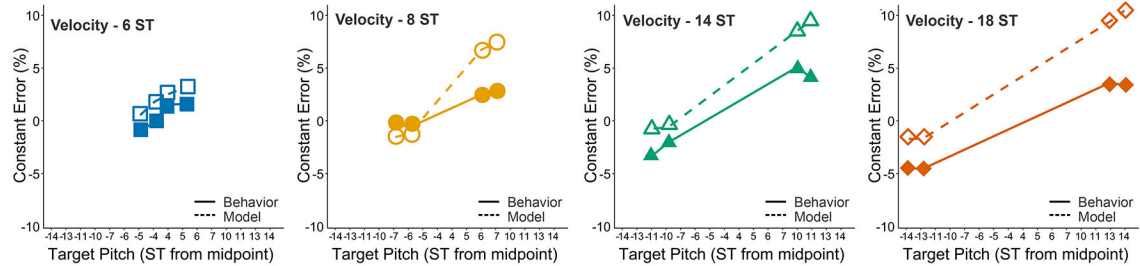
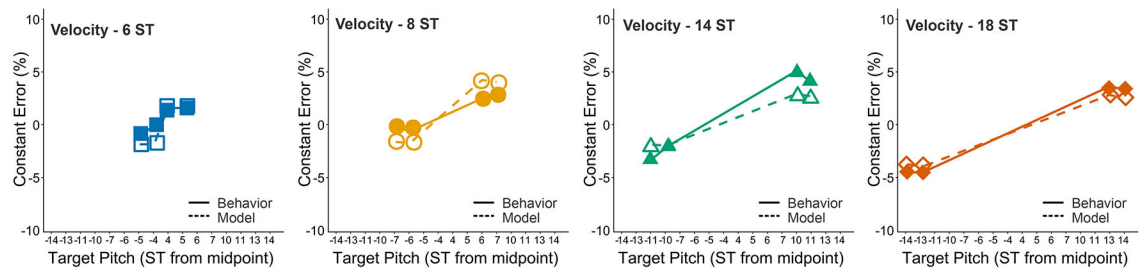


Figure 5.

Experiment 2 results showing mean CE with standard error bars as a function of target relative pitch condition with line of best fit. Negative values are X pitch closer to A and positive values are closer to B. The positive slope represents the Kappa effect observed for inconsistent pitch-trajectory sequences.

A Imputed velocity model**B Grouping model****Figure 6.**

Average CE from Experiment 2 showing observed data in solid lines with filled markers and model-predicted data in dashed lines with open markers. **A)** Imputed velocity model. **B)** Perceptual grouping model. Best-fitting model parameters were estimated for each subject and pitch-velocity, then respective models were used to calculate predicted proportion of “long-short” responses (Equation 2), and CEs were estimated following the procedures detailed above.